

Lecture 13

Patterns

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- Generally, patterns are regular occurrences of an object in some given space/dimension.
 - ‘Architectural patterns’ are solutions to problems belonging to some family of other, similar, architectural patterns.
- An ‘architectural style’ is a set of characteristics and features which distinguish architectural designs from one another.
- The repeated use of certain patterns defines a software’s architectural style.
 - A given piece of software may be composed of multiple different architectural styles.
- An architecture’s ‘orientation’ the kinds of concerns it is focused on addressing.
- ‘Interaction-oriented’ software architectures focus on challenges associated with:
 1. Components interacting with other components.
 2. Components interacting with the environment.
- Some examples of interaction-oriented software include PAC and MVC.
 - Both will be explained in detail in later lectures.
- Interaction-oriented architectures segregate a system’s components into three major modules:
 - A data module.
 - A flow control module.
 - A view/presentation module.
- Systems with high coupling generally:
 - Make modifying the system more difficult.
 - Are better at separating out concerns.